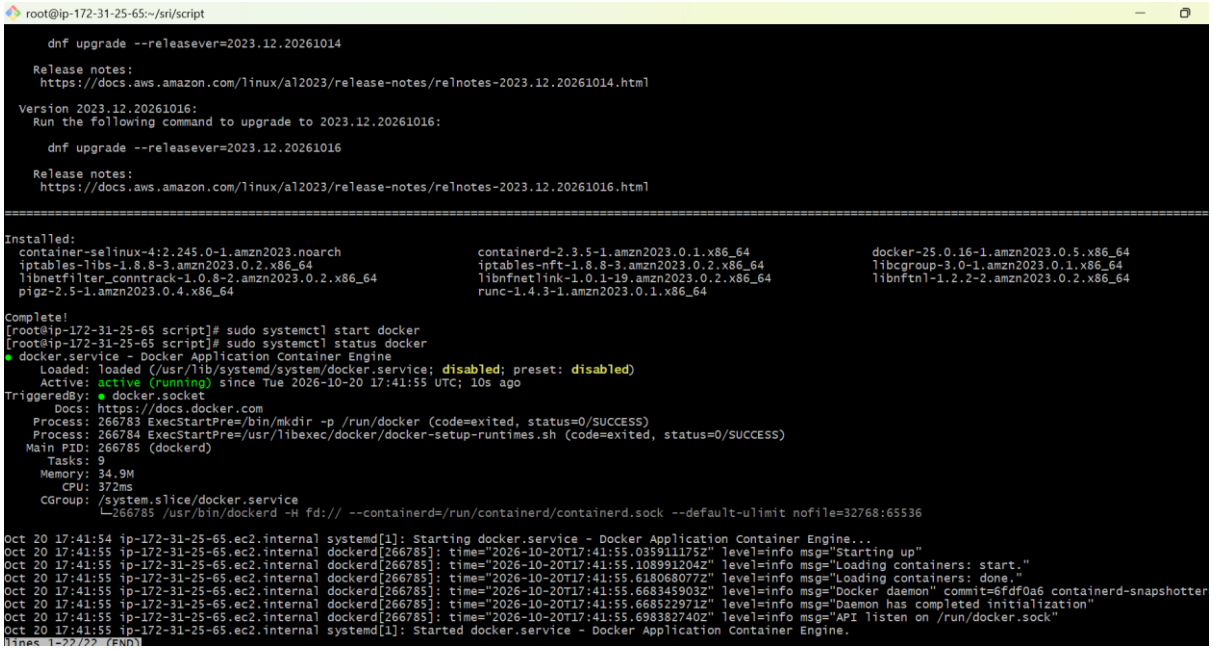
Linux Challenge

1. Install Docker in AWS EC2 server.



1. Update packages and install Docker: **sudo dnf update -y**

sudo dnf install docker -y

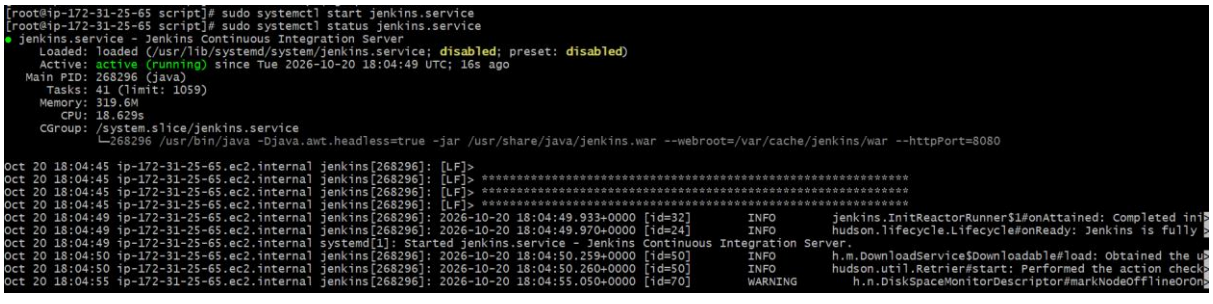
2. Start Docker: **sudo systemctl start docker**

3. Enable Docker at startup: **sudo systemctl enable docker**

4. Verify Docker installation: **sudo systemctl status docker**

sudo docker run hello-world

2 Install Jenkins in Aws EC2



Jenkins is an automation tool used in DevOps to build, test, and deploy applications automatically.

1. Updated the system

Command: **sudo dnf update -y**

Explanation: Updates the packages on your Amazon Linux EC2 instance so the system has the latest available updates.

2. Install Java

Command: **sudo dnf install java-21-amazon-corretto -y**

Explanation: Jenkins runs on Java, so we install Java before installing Jenkins.

3. Download the Jenkins repository and key

Commands:

- **sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/rpm-stable/jenkins.repo**

- **sudo rpm --import https://pkg.jenkins.io/rpm-stable/jenkins.io-2026.key**

Explanation: The repository tells Amazon Linux where to find Jenkins packages. The key helps verify that the packages are trusted.

4. Install Jenkins

Command: **sudo dnf install jenkins -y**

Explanation: Downloads and installs the Jenkins software on your EC2 instance.

5. Start Jenkins and enable automatic startup

Commands:

- **sudo systemctl start jenkins**

- sudo systemctl enable jenkins

Explanation: The first command starts Jenkins now. The second makes it start automatically when the EC2 instance reboots.

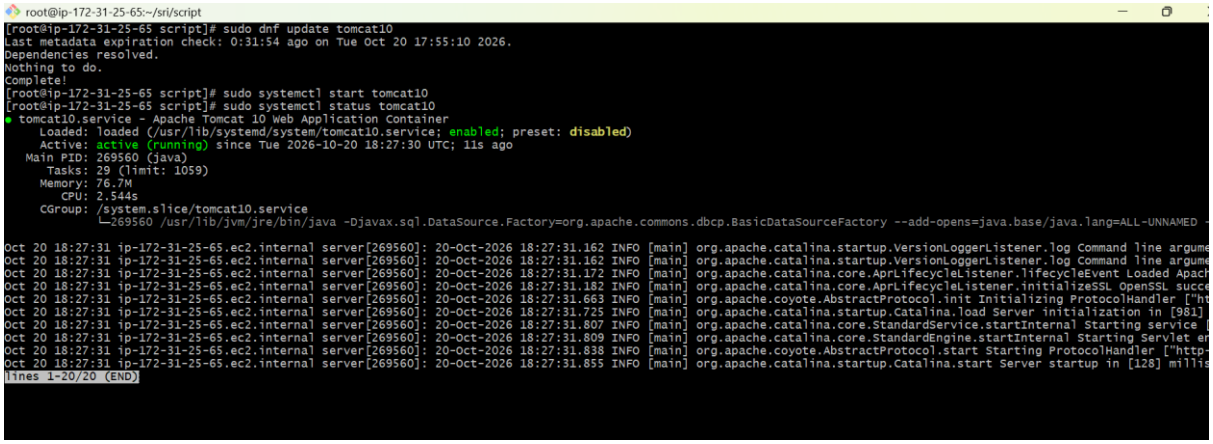
6. Access Jenkins in your browser

First, allow inbound TCP port 8080 in your EC2 security group, if that port is available. Then open:

Explanation: This opens the Jenkins web interface. Complete the initial setup using the administrator password from:

sudo cat /var/lib/jenkins/secrets/initialAdminPassword

3 Install Apache Tomcat10 in AWS EC2



- To install apache tomcat10 we need to install java development kit (jdk 17) first.
- After that the apache tomcat10 version need to be installed.
- Using the cmd **sudo dnf install tomcat10**.
- After installing the process, we should start the process by cmd **sudo systemctl start tomcat10** which listens in port :8080.
- Therefore, later check the status of the tomcat10. By **sudo systemctl status tomcat10**.